Anas Ahmad

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.04
Little Flower School	Class XII (Council for the Indian School Certificate Examinations (CISCE))	2019	88.20%
Little Flower School	Class X (Council for the Indian School Certificate Examinations (CISCE))	2017	87.60%

INTERNSHIPS

IIT Kanpur | Project Intern

- **Characterized ionic conductivity** of composite solid-state electrolytes at six dispersion ratios (10%, 20%, 30%, 40%, 50%, 60%), using **Origin** for data analysis and plotting to establish composition-conductivity trends for electrolyte optimization.
- **Assembled coin/test cells** inside a vacuum glove-box chamber for a range of **electrolyte formulations**, ensuring moisture- and oxygen-free processing

Metcalf house (DRDO) | Project Intern

- **Implemented UART communication protocol** on FPGA, designing transmitter and receiver modules in **Verilog/VHDL.**

PROJECTS

Wireless Power Transfer System for EV Charging | M.tech Thesis

- **Designed and experimentally validated a 2.2kW hardware prototype** integrating a dual sided LCC compensated network, full bridge converter, gate drivers, sensing circuits and magnetic couplers; achieved 77.07% peak efficiency, demonstrating practical feasibility of the proposed WPT architecture.
- **Developed fixed-frequency primary-side phase-shift control with PI regulation** to achieve stable output-voltage tracking under load and coupling variations, including disturbance rejection during coil misalignment; experimentally validated ZVS operation of the primary SiC MOSFET inverter.
- **Extending the WPT system toward 7.7 kW** operation through hardware design and experimental testing, with ongoing development of a **Current-Fed Inverter (CFI)** integration to **enhance high-power wireless charging** performance.

Real-Time Power Quality Monitoring and Harmonic Analysis

- **Engineered a TI C2000-based real-time power-quality monitoring platform** for synchronized voltage/current sampling and digital computation of RMS, frequency, power factor, and THD.
- **Implemented Goertzel algorithm** for harmonic spectrum extraction and power-quality assessment, enabling real-time detection of harmonic components and waveform distortions.
- Integrated **Blynk Cloud for real-time monitoring** of power-quality parameters.

Design and Digital Control of 200 W DC-DC Boost Converter Using TI C2000

- **Achieved tight output-voltage regulation** by implementing a discrete **PI compensator** on a **TI C2000** MCU in **embedded C**, driving **ePWM** duty from **ADC** feedback in an **ISR** -- holding the setpoint under line and load steps.
- **Delivered a low-noise 200 W power board** by designing the **2-layer PCB in KiCad** with minimized gate-drive loops, star grounding, and wide current paths -- enabling clean sensing and stable full-load operation.

Real Time Monitoring of a Single Axis Solar Tracking System Using NI LabView | B.tech Thesis

- **Analyzed** six DC-DC converter topologies (buck, boost, buck-boost, split-pi, flyback, linear regulator) for a solar PV system using **MATLAB/Simulink** modeling, identifying the **split-pi converter** as most efficient with lowest voltage ripple.
- **Built** a **LabVIEW**-based real-time monitoring system for a single-axis solar tracker with a **buck-boost converter** (used for hardware demo due to design constraints), logging voltage, current, and power via **DAQ** for fault detection.
- **Optimized** MPPT-based power extraction by mathematically modeling converter duty ratios, improving output power reliability and efficiency of the solar PV system.

Robotic Arm with Pen-Holding Mechanism | Personal Project

- **Designed and built** a robotic arm using multiple 1A-rated servo motors interfaced with an **Arduino UNO** for multi-axis pen-holding and manipulation.
- **Programmed** servo control logic in embedded C, achieving smooth and precise coordinated motion for accurate pen positioning.

Modelling and closed loop simulation of a three phase bi-directional voltage source converter (VSC) | Personal Project

- **Simulated a 10 kVA bidirectional three-phase grid-connected VSC** in MATLAB/Simulink using dq0-based current control and sine-triangle PWM, achieving unity power factor across rectifier/inverter modes.
- **Designed PI current controllers via pole-zero cancellation** with filter sizing and cross-coupling decoupling, achieving <5 ms settling time and <5% overshoot.

SKILLS

KiCad, OrCaD, Altium, Expert in MATLAB, Python, C programming, Originlab, LabView

POSITIONS OF RESPONSIBILITY

Volunteer, ICPS 2025 (International Conference on Power Systems), IIT Hyderabad -- supported conference operations for an international gathering of power systems researchers

PUBLICATIONS

- Performance Analysis of Solar PV System for different configurations, Journal Of Physics : Conference Series · Aug 31, 2023

- Performance Analysis of Solar PV System integrated with SEPIC and ZETA converters, Journal of Physics : Conference Series · Aug 31, 2023

AWARDS AND RECOGNITIONS

- 1st Place, Electrical Electronics domainScrolls'23 national technical paper presentation, AKGEC (May 2023).

HOBBIES AND INTEREST

- Self trained dancer (hip-hop, free style)
- Video games like valorant, counter strike, AAA games
- Love to read thriller novels
- Badminton